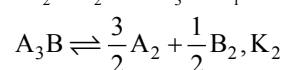
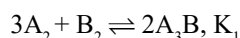


CHAPTER 6

Equilibrium

Law of Chemical Equilibrium, Equilibrium Const. and Its Application

1. At a given temperature and pressure, the equilibrium constant values for the equilibria are given below: (2024 Re)



The relation between K_1 and K_2 is:

a. $K_1^2 = 2K_2$

b. $K_2 = \frac{K_1}{2}$

c. $K_1 = \frac{1}{\sqrt{K_2}}$

d. $K_2 = \frac{1}{\sqrt{K_1}}$

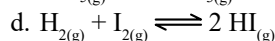
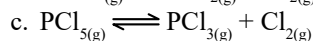
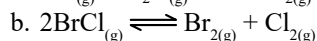
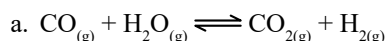
2. For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is: (2024)

$$[A] = [B] = [C] = 2 \times 10^{-3} \text{ M.}$$

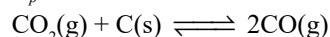
Then, which of the following is correct?

- Reaction has a tendency to go in backward direction.
- Reaction has gone to completion in forward direction.
- Reaction is at equilibrium.
- Reaction has a tendency to go in forward direction.

3. In which of the following equilibria, K_p and K_c are **NOT** equal? (2024)



4. K_p for the following reaction is 3.0 at 1000 K. (2022 Re)



What will be the value of K_c for the reaction at the same temperature?

(Given – $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)

a. 3.6

b. 0.36

c. 3.6×10^{-2}

d. 3.6×10^{-3}

5. $3\text{O}_2(g) \rightleftharpoons 2\text{O}_3(g)$ for the given reaction at 298 K, K_c is found to be 3.0×10^{-59} . If the concentration of O_2 at equilibrium is 0.040 M, then concentration of O_3 in M is: (2022)

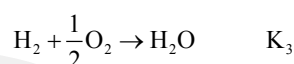
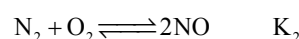
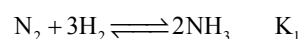
a. 1.2×10^{21}

b. 4.38×10^{-32}

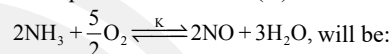
c. 1.9×10^{-63}

d. 2.4×10^{31}

6. The equilibrium constants of the following are:



The equilibrium constant (K) of the reaction: (2017-Delhi)



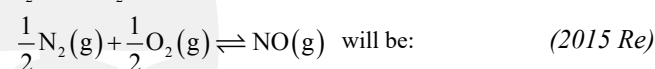
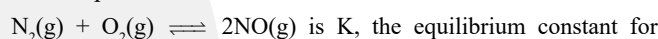
a. $K_2^3 K_3 / K_1$

b. $K_1 K_3^3 / K_2$

c. $K_2 K_3^3 / K_1$

d. $K_2 K_3 / K_1$

7. If the equilibrium constant for



a. K^2

b. $K^{1/2}$

c. $\frac{1}{2}K$

d. K

8. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12} , then at equilibrium the system will contain: (2015)

a. Mostly reactants

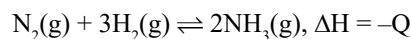
b. Mostly products

c. Similar amounts of reactants and products

d. All reactants

Relation Between K, Q and G , Factors Affecting Equilibria

9. For the reaction in equilibrium (2024 Re)



Reaction is favoured in forward direction by:

a. use of catalyst

b. decreasing concentration of N_2

c. low pressure, high temperature and high concentration of ammonia

d. high pressure, low temperature and higher concentration of H_2

10. The equilibrium concentrations of the species in the reaction $A + B \rightleftharpoons C + D$ are 2, 3, 10 and 6 mol L^{-1} respectively at 300 K. ΔG° for the reaction is (2023)

($R = 2 \text{ cal/mol K}$)

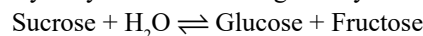
a. -13.73 cal

b. 1372.60 cal

c. -137.26 cal

d. -1381.80 cal

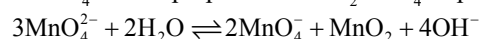
11. Hydrolysis of sucrose is given by the following reaction:



If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^\circ$ at the same temperature will be : (2020)

- $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (2 \times 10^{13})$
 - $8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (3 \times 10^{13})$
 - $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (4 \times 10^{13})$
 - $-8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (2 \times 10^{13})$
12. Which one of the following conditions will favour maximum formation of the product in the reaction,
 $\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons \text{X}_2(\text{g}) \quad \Delta_r H = -X \text{ kJ}$ (2018)
- Low temperature and high pressure
 - Low temperature and low pressure
 - High temperature and low pressure
 - High temperature and high pressure
13. Which one of the following statements is not correct? (2017-Delhi)
- Coenzymes increase the catalytic activity of enzyme
 - Catalyst does not initiate any reaction
 - The value of equilibrium constant is changed in the presence of a catalyst in the reaction at equilibrium
 - Enzymes catalyse mainly bio-chemical reactions
14. A 20 litre container at 400 K contains $\text{CO}_2(\text{g})$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the containers is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be: (2017-Delhi)
- (Given that: $\text{SrCO}_3(\text{s}) \rightleftharpoons \text{SrO}(\text{s}) + \text{CO}_2(\text{g}) \quad K_p = 1.6 \text{ atm}$)
- 2 litre
 - 5 litre
 - 10 litre
 - 4 litre
15. Consider the nitration of benzene using mixed conc. H_2SO_4 and HNO_3 . If a larger amount of KHSO_4 is added to the mixture, the rate of nitration will be: (2016 - I)
- Doubled
 - Increase
 - Decrease
 - Unchanged
16. Which of the following statements is correct for a reversible process in a state of equilibrium? (2015)
- $\Delta G^\circ = -2.303 \text{ RT log } K$
 - $\Delta G^\circ = 2.303 \text{ RT log } K$
 - $\Delta G = -2.303 \text{ RT log } K$
 - $\Delta G = 2.303 \text{ RT log } K$
17. For a given exothermic reaction, K_p and K'_p are the equilibrium constants at temperature T_1 and T_2 , respectively. Assuming that heat of reaction is constant in temperature range between T_1 and T_2 , it is readily observed that (2014)
- $K_p > K'_p$
 - $K_p < K'_p$
 - $K_p = K'_p$
 - $K_p = \frac{1}{K'_p}$
18. For the reversible reaction,
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{Heat}$. The equilibrium shifts in forward direction: (2014)
- By decreasing the pressure
 - By decreasing the concentrations of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$
 - By increasing pressure and decreasing temperature
 - By increasing the concentration of $\text{NH}_3(\text{g})$

19. KMnO_4 can be prepared from K_2MnO_4 as per the reaction:



The reaction can go to completion by removing OH^- ions by adding: (2013)

- HCl
- KOH
- CO_2
- SO_2

Acids, Bases and Salts

20. Conjugate base for Bronsted acids H_2O and HF are: (2019)

- OH^- and H_2F^+ , respectively
- H_3O^+ and F^- , respectively
- OH^- and F^- , respectively
- H_3O^+ and H_2F^+ , respectively

21. Which of the following fluoro-compounds is most likely to behave as a Lewis base? (2016 - II)

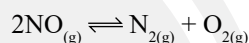
- CF_4
- SiF_4
- BF_3
- PF_3

22. Which of these is least likely to act as a Lewis base? (2013)

- CO
- F^-
- BF_3
- PF_3

Ionization of Acids & Bases, pH Scale, Hydrolysis

23. Consider the following reaction in a sealed vessel at equilibrium with concentrations of $\text{N}_2 = 3.0 \times 10^{-3} \text{ M}$, $\text{O}_2 = 4.2 \times 10^{-3} \text{ M}$ and $\text{NO} = 2.8 \times 10^{-3} \text{ M}$. (2024)



If 0.1 mol L^{-1} of $\text{NO}_{(\text{g})}$ is taken in a closed vessel, what will be degree of dissociation (α) of $\text{NO}_{(\text{g})}$ at equilibrium ?

- 0.8889
- 0.717
- 0.00889
- 0.0889

24. For a weak acid HA, the percentage of dissociation in nearly 1% at equilibrium. If the concentration of acid is 0.1 mol L^{-1} , then the **correct** option for its K_a at the same temperature is: (2023-Manipur)

- 1×10^{-4}
- 1×10^{-6}
- 1×10^{-5}
- 1×10^{-3}

25. 0.01 M acetic acid solution is 1% ionised, then pH of this acetic acid solution is: (2022 Re)

- 1
- 3
- 2
- 4

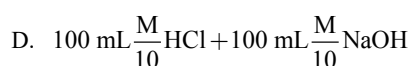
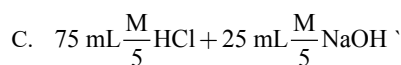
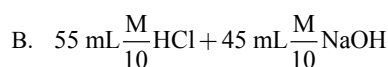
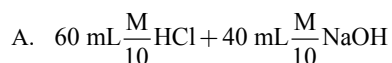
26. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethyl ammonium acetate solution is: (2021)

- 5.50
- 7.75
- 6.25
- 8.50

27. Which among the following salt solutions is **basic** in nature? (2020-Covid)

- Ammonium sulphate
- Ammonium nitrate
- Sodium acetate
- Ammonium chloride

28. Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations: (2018)



pH of which one of them will be equal to 1?

- a. B b. Ac. c. d. D

29. The percentage of pyridine (C_5H_5N) that forms pyrimidine ion ($C_5H_5N^+H$) in a 0.10 M aqueous pyridine solution (K_b for $C_5H_5N = 1.7 \times 10^{-9}$) is: (2016 - II)
a. 0.77% b. 1.6% c. 0.0060% d. 0.013%
30. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed? (2015 Re)
a. 2.0 b. 7.0 c. 1.04 d. 12.65
31. Aqueous solution of which of the following compounds is the best conductor of electric current? (2015 Re)
a. Hydrochloric acid, HCl b. Ammonia, NH_3
c. Fructose, $C_6H_{12}O_6$ d. Acetic acid, $C_2H_4O_2$
32. Which of the following salts will give highest pH in water? (2014)
a. NaCl b. Na_2CO_3 c. $CuSO_4$ d. KCl

Buffer Solutions

33. An acidic buffer is prepared by mixing: (2023-Manipur)
a. weak acid and its salt with strong base
b. equal volume of equimolar solutions of weak acid and weak base.
c. strong acid and its salt with strong base
d. strong acid and its salt with weak base
(The pK_a of acid = pK_b of the base)
34. The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is (2022)
[Given pK_a of $CH_3COOH = 4.57$]
a. 2.57 b. 5.57 c. 3.57 d. 4.57
35. Which will make basic buffer? (2019)
a. 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH_3COOH
b. 100 mL of 0.1 M CH_3COOH + 100 mL of 0.1 M NaOH
c. 100 mL of 0.1 M HCl + 200 mL of 0.1 M NH_4OH
d. 100 mL of 0.1 M HCl + 100 mL of 0.1 M NaOH
36. Which one of the following pairs of solution is not an acidic buffer? (2015 Re)
a. H_3PO_4 and Na_3PO_4
b. $HClO_4$ and $NaClO_4$
c. CH_3COOH and CH_3COONa
d. H_2CO_3 and Na_2CO_3

Solubility Equilibria of Sparingly Soluble Salts

37. The ratio of solubility of AgCl in 0.1 M KCl solution to the solubility of AgCl in water is:
(Given : Solubility product of AgCl = 10^{-10}) (2024 Re)
a. 10^{-4} b. 10^{-6} c. 10^{-9} d. 10^{-5}
38. Find out the solubility of $Ni(OH)_2$ in 0.1 M NaOH. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15} . (2020)
a. 2×10^{-8} M b. 1×10^{-13} M
c. 1×10^8 M d. 2×10^{-13} M

39. HCl was passed through a solution of $CaCl_2$, $MgCl_2$ and NaCl. Which of the following compounds crystallizes? (2020)
a. Only NaCl b. Only $MgCl_2$
c. NaCl, $MgCl_2$ and $CaCl_2$ d. Both $MgCl_2$ and $CaCl_2$
40. The solubility product for a salt of the type AB is 4×10^{-8} . What is the molarity of its standard solution? (2020-Covid)
a. 16×10^{-16} mol/L b. 2×10^{-16} mol/L
c. 4×10^{-4} mol/L d. 2×10^{-4} mol/L
41. pH of a saturated solution of $Ca(OH)_2$ is 9. The solubility product (K_{sp}) of $Ca(OH)_2$ is: (2019)
a. 0.5×10^{-15} b. 0.25×10^{-10}
c. 0.125×10^{-15} d. 0.5×10^{-10}
42. The solubility of $BaSO_4$ in water is 2.42×10^{-3} g L^{-1} at 298 K. The value of its solubility product (K_{sp}) will be (2018)
(Given molar mass of $BaSO_4 = 233$ g mol^{-1})
a. 1.08×10^{-10} $mol^2 L^{-2}$ b. 1.08×10^{-12} $mol^2 L^{-2}$
c. 1.08×10^{-8} $mol^2 L^{-2}$ d. 1.08×10^{-14} $mol^2 L^{-2}$
43. Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is 2.2×10^{-4} mol L^{-1} . Solubility product of $Ag_2C_2O_4$ is: (2017-Delhi)
a. 5.3×10^{-12} b. 2.42×10^{-8}
c. 2.66×10^{-12} d. 4.5×10^{-11}
44. The solubility of $AgCl(s)$ with solubility product 1.6×10^{-10} in 0.1 M NaCl solution would be: (2016 - II)
a. 1.6×10^{-11} M b. Zero
c. 1.26×10^{-5} M d. 1.6×10^{-9} M
45. MY and NY_3 , two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature, which statements would be true in regard to MY and NY_3 ? (2016 - I)
a. The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities
b. The molar solubilities of MY and NY_3 in water are identical
c. The molar solubility of MY in water is less than that of NY_3
d. The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water
46. The K_{sp} of Ag_2CrO_4 , AgCl, AgBr and AgI are respectively, 1.1×10^{-12} , 1.8×10^{-10} , 5.0×10^{-13} , 8.3×10^{-17} . Which one of the following salts will precipitate last if $AgNO_3$ solution is added to the solution containing equal moles of NaCl, NaBr, NaI and Na_2CrO_4 ? (2015)
a. AgCl b. AgBr c. Ag_2CrO_4 d. AgI
47. Using the Gibb's energy change, $\Delta G^\circ = + 63.3$ kJ, for the following reaction,
 $Ag_2CO_3(s) \rightleftharpoons 2Ag^+(aq) + CO_3^{2-}(aq)$
the K_{sp} of $Ag_2CO_3(s)$ in water at 25°C is ($R = 8.314$ J $K^{-1} mol^{-1}$) (2014)
a. 3.2×10^{-26} b. 8.0×10^{-12}
c. 2.9×10^{-3} d. 7.9×10^{-2}
48. Identify the correct order of solubility in aqueous medium: (2013)
a. $Na_2S > CuS > ZnS$ b. $Na_2S > ZnS > CuS$
c. $CuS > ZnS > Na_2S$ d. $ZnS > Na_2S > CuS$

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (c) | 4. (c) | 5. (b) | 6. (c) | 7. (b) | 8. (b) | 9. (d) | 10. (d) |
| 11. (d) | 12. (a) | 13. (c) | 14. (b) | 15. (c) | 16. (a) | 17. (a) | 18. (c) | 19. (c) | 20. (c) |
| 21. (d) | 22. (c) | 23. (b) | 24. (c) | 25. (d) | 26. (b) | 27. (c) | 28. (c) | 29. (d) | 30. (d) |
| 31. (a) | 32. (b) | 33. (a) | 34. (b) | 35. (c) | 36. (b) | 37. (a) | 38. (d) | 39. (a) | 40. (d) |
| 41. (a) | 42. (a) | 43. (a) | 44. (d) | 45. (c) | 46. (c) | 47. (b) | 48. (b) | | |

